SAC 5: Bio-organics and Chromatography **Name: _____**

Teacher: MC, RH, SP, NP, JJ (please circle) **Total marks 45**

You will be provided with data booklet. Please do not write in it. You may use a scientific calculator only. All mobile phones must be placed on the front bench. No reading time is available

Section A. For the following questions, circle the letter corresponding to the best alternative.

Question 1

When Thin Layer Chromatography is used, one substance with an R_f of 0.70 travels 5.6 cm from the origin. How far has the solvent front travelled from the origin under the same conditions?

A 3.9 cm **B** 5.6 cm

C 7.0 cm **D** 8.0 cm

Question 2

As an amino acid, glycine, dissolves in water. In an aqueous solution with a pH of 12.0, glycine is likely to act as

A an acid only **B** a base only

C neither and acid or base **D** both an acid and a base

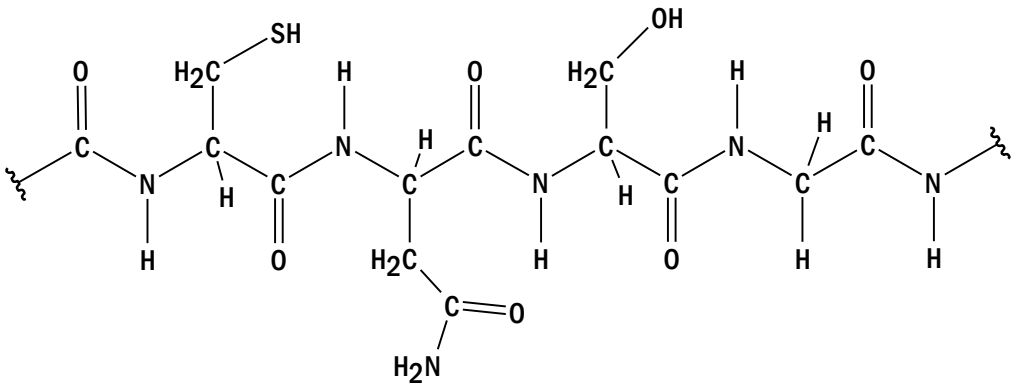
Question 3

During the condensation reactions that form proteins, triglycerides, and polysaccharides, links are formed between the reactants. Which alternative shows the correct links in order for these 3 molecules?

A ester, ether, amide **B** amide, ester, ether

C ether, amide, ester **D** amide, ether, ester

Refer to the diagram below to answer Questions 4 and 5



Question 4

Identify the amino acid that is **NOT** represented in this section of protein chain

A Asparagine **B** Aspartic Acid

C Cysteine **D** Serine

Question 5

Which of the amino acid residues in the chain is likely to form a covalent bond with another protein chain?

A Glycine **B** Aspartic Acid

C Cysteine **D** Serine

Question 6

Which of the following is most likely to be found in body fat?

A Vitamin D_2 **B** Glycerol

C Lactose **D** Vitamin C

Question 7

Some triglycerides are found as solids at room temperature while others are liquid. Choose the best statement to account for this difference

A Saturated fats tend to be liquids because their single carbon to carbon bonds prevent their molecules packing as closely together.

B Polyunsaturated fats tend to be liquid as their rigid carbon to carbon double bonds prevent their molecules packing closer together.

C Saturated fats tend to be solid because their molecules are shorter and so they can pack closer together.

D Polyunsaturated fats tend to be solid because their molecules have greater polarity and so bond together more readily.

Questions 8 and 9 refer to tripeptides formed from two glycine molecules and one alanine molecule . $M_{Glycine}$ = 75.0 g mol^{-1}, $M_{Alanine}$ = 89.0 g mol^{-1}.

Question 8

What is the molar mass of any tripeptide formed from two glycine and one alanine amino acids?

A 185 g mol^{-1}

B 203 g mol^{-1}

C 239 g mol^{-1}

D The molar mass depends on the order the three amino acids are joined.

Question 9

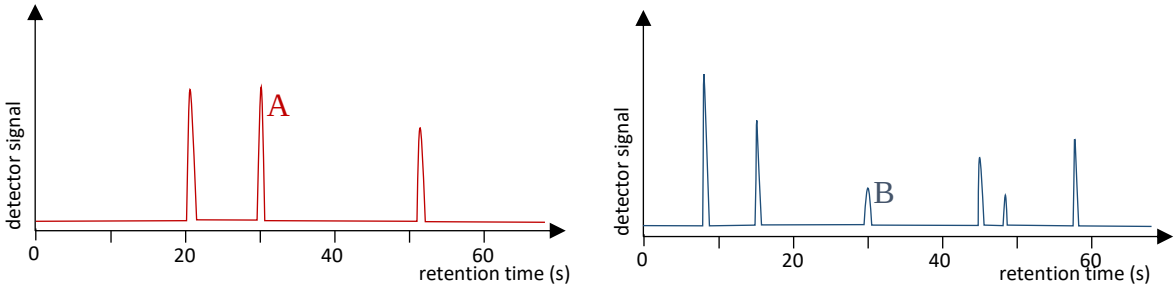
How many different structural isomers of two glycine and one alanine can be formed?

A 2 **B** 3

C 4 **D** 6

Question 10

The chromatogram below identifies peak A and peak B. Analysis of these peaks would be:



A. Peak A and B are for the same chemical, but the quantity of A is greater.

B. Peak A and B are for the same chemical, but the quantity of B is greater.

C. Peak A and B are the same amount, but A is more strongly attracted to the stationary phase than B.

D. Peak A and B are the same amount, but B is more strongly attracted to the stationary phase than A.

Question 11

Which of the following is an omega-3 fatty acid?

A palmitic acid

B oleic acid

C linolenic acid

D arachidonic acid

Question 12

The following substances are passed through an HPLC column with a non-polar stationary phase and their retention times recorded. Which is likely to have the shortest retention time?

A butan-1-ol

B butan-1-amine

C butanal

D butanoic acid

Question 13

Rancidity in foods refers to a notable deterioration in taste or smell. One such process is called oxidative rancidity. It is likely to occur most quickly for

A short chained fats

B polyunsaturated fats

C long chained protein molecules

D polysaccharides

Question 14

Coenzymes are molecules that

A are the starting material in the production of protein from amino acids.

B enzymes that work in pairs to get a particular job done.

C non-protein compound that some enzymes need to be active.

D protein materials that react with enzymes to produce kinetic energy.

Question 15

The secondary structure of a protein refers to

A the sequence of amino acids bonded together by peptide links

B the folding resulting from the formation of interactions between side groups

C the formation of coils and sheets resulting from hydrogen-bonding between distant peptide links.

D the folding resulting from the formation of disulfide links.

(15 marks)

END OF SECTION A

Section B **Answer all question**

Question 1

a) Food safety guidelines state that left over food should be stored in the refrigerator and, when reheated, heated above 70 ^{0}C for 5 minutes prior to serving. Explain why each process slows down reactions involving enzymes in the microbes that cause spoilage.

i) Cooling

_____

_____

_____

ii) Heating

_____

_____

_____

b) During the digestion of food, renin is an enzyme that helps curdle milk in the stomach. How might a student test that **cooling** has prevented any reaction in the way described above?

_____

_____

_____

(1 + 1 + 2 = 4 marks)

Question 2

a) What type of fatty acid is arachidonic acid? Circle the correct term below:

Saturated **Mono-unsaturated** **Polyunsaturated**

b) If two mol of arachidonic acid were reacted with bromine (Br_2), what mass of bromine would be expected to readily react?

_____

_____

_____

c) Draw a semi-structural diagram of the triglyceride from which **stearic acid** would be derived.

d) Give the molecular formulae of the products formed from the hydrolysis reaction of the triglyceride you have drawn in c).

_____

(1 + 2 + 2 + 2 = 7 marks)

Question 3

a) When leucine (Leu) reacts with alanine (Ala), two dipeptides can be formed. Write the **semi-structural** formula of **both** of these dipeptides **<u>and</u>** clearly circle and name the link.

_____

_____

b) **Name** the substance formed in addition to the dipeptide in the reaction described above.

_____

c) The function of a protein is dependent on its three-dimensional structure. This structure can be disrupted, denaturing the protein. Which of the following changes could cause this denaturing?

I	the addition of a strong acid
II	the addition of a strong base
III	a significant increase in temperature

A. I only

B. I and II only

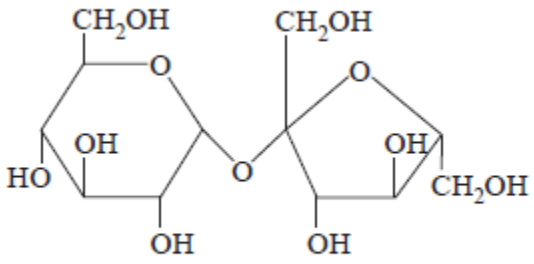
C. III only

D. I, II and III

(3 + 1 + 1 = 5 marks)

Question 4

The molecule sucrose is shown below.



sucrose

a) Sucrose undergoes enzyme catalysed hydrolysis during digestion. Draw the structure of one of the products of this hydrolysis **and name the molecule**.

Name: _____

b) A bomb calorimeter is used to determine the energy content of sucrose. 8.48 g of sucrose is completely combusted in a calorimeter containing 2.00 L of water. The temperature of the water rose from 20.2 ^{0}C to 35.4 ^{0}C. Calculate the energy content of sucrose in **kJ g^{-1}**.

_____

_____

_____

c) Determine the heat energy per mol of solid sucrose and hence write a thermochemical equation for the complete combustion of sucrose.

_____

_____

_____

(2 + 2 + 3 = 7 marks)

Question 5

The speed at which a food increases the blood's sugar levels is an important consideration both for athletes and the general public. Glucose is given a glycaemic index (GI) of 100 while wholemeal bread has a GI of approximately 59.

a) Explain why glucose is assigned a GI of 100.

_____

b) Bread brand A was made from flour with a high ratio of amylose to amylopectin, while in brand B the amylose content was lower. Which brand is likely to have a higher GI and why?

_____

_____

_____

c) Bicycle riders going on a long ride often consume foods like bananas before and along the way. Explain why this is preferred to other snacks, like lollies, for energy.

_____

_____

_____

(1 + 2 + 2 = 5 marks)

Question 6

Oxidative rancidity leads to the formation of unpleasant smelling and potentially harmful chemicals. Give two ways rancidity can be reduced.

i) _____

ii) _____

(2 marks)

End of SAC Total Marks 45